

AI-Driven Parking Intelligence: Detecting Illegal Hotspots and Quantifying Congestion Impact

1. Feasibility Assessment

Technical Feasibility of Detection Modalities

The technical feasibility of deploying artificial intelligence to detect illegal and spillover parking relies heavily on the sensory modalities utilised to capture the urban environment. An exhaustive analysis of available technologies reveals distinct operational advantages and inherent limitations for each modality in the context of continuous enforcement.

Fixed CCTV and traffic surveillance cameras remain the most technically viable and scalable modality for detecting illegal parking. Advanced computer vision architectures, particularly single-stage object detectors such as YOLOv8 and YOLOv11, demonstrate exceptional real-time accuracy in localising stationary vehicles within predefined polygonal regions of interest (ROIs)¹. These systems can seamlessly integrate with existing municipal IP camera networks without requiring massive new hardware investments³. However, their fixed nature leaves them vulnerable to occlusions caused by larger transit vehicles or urban infrastructure. Mobile and dashcam footage offers a dynamic, high-coverage alternative that bypasses the fixed-infrastructure cost. Recent academic frameworks have successfully utilised two-stage pseudo-labelling pipelines applied to dashcam imagery. By deploying instance segmentation models to detect vehicles and semantic road markings (such as red no-parking lines), spatial analysis algorithms can compute the exact pixel distance between the vehicle mask and the restricted boundary, successfully classifying illegal parking from a moving vantage point¹. Nevertheless, mobile detection suffers from temporal sparsity; it cannot reliably track the continuous dwell time of a parked vehicle unless the patrol route is highly repetitive.

Drone and aerial imagery provide unparalleled bird's-eye perspectives, entirely eliminating the occlusion problems endemic to street-level CCTV. Datasets such as highD and DroneVehicle demonstrate that aerial platforms equipped with dual RGB and Infrared (IR) sensors can achieve highly robust cross-modality detection, even under extremely low-light conditions⁴. Despite these algorithmic successes, the technical feasibility of drones for continuous municipal enforcement is negated by severe operational constraints, including restricted battery life, stringent urban airspace regulations, and susceptibility to adverse weather. Similarly, satellite imagery lacks the spatial resolution and temporal frequency required for real-time, dynamic enforcement, rendering it suitable only for macro-level, historical urban planning.

Internet of Things (IoT) ground sensors, specifically triaxial magnetic and ultrasonic sensors, present a highly accurate deterministic approach to occupancy detection. Operating over low-power wide-area networks (LoRaWAN), these sensors identify the presence of a metal

mass or ultrasonic echo alterations with near-perfect accuracy⁷. While highly feasible for detecting occupancy, ground sensors suffer from a critical technical deficit: they cannot classify the vehicle type, read license plates, or distinguish between a legally parked vehicle and a violating one without secondary optical integration.

Crowdsourced reporting applications leverage ubiquitous smartphone penetration to capture acute, hyper-localised violations that evade fixed networks. Feasibility studies indicate that participatory platforms—such as those designed to report micromobility infractions or via Web-GIS systems—can supply high-quality, geo-tagged photographic evidence¹⁰. However, crowdsourcing introduces severe spatial bias, as reporting frequency correlates with civic engagement rather than objective violation density, necessitating rigorous algorithmic filtering to eliminate spam¹².

GPS and fleet telematics data provide rich, granular insights into vehicular dwell times and intersection traversal speeds. While telematics can flawlessly pinpoint the location of commercial or transit fleets, it captures only a fraction of private commuter traffic. Consequently, relying solely on telematics fails to identify the vast majority of illegal parking incidents perpetrated by private citizens. Furthermore, parking meter and sensor API data can indicate payment violations but are structurally blind to off-bay spillover parking on carriageways and pavements.

Compute and Latency Feasibility: Edge vs. Cloud Architecture

The architectural dichotomy between edge inference and cloud-based batch processing dictates the latency, bandwidth, and operational expenditure of the intelligence system. Streaming continuous high-definition video feeds to a centralised cloud server for detection is computationally prohibitive at a city-wide scale. Empirical estimates indicate that streaming 1920×1080 resolution footage from a single camera can result in the transfer of up to 35 gigabytes of data per hour¹³.

Edge AI inference has proven highly feasible and is increasingly adopted as the industry standard. By deploying lightweight neural networks, such as MobileNetV3 or quantised YOLO variants, directly on edge-compute nodes (e.g., NVIDIA Jetson or dedicated AI processors), the video is processed locally¹⁴. The edge device extracts bounding boxes, performs optical character recognition (OCR) on license plates, and tracks dwell times. It then transmits only a lightweight payload of alphanumeric metadata and isolated image crops to the cloud¹⁶. This architecture reduces network latency to mere milliseconds, virtually eliminates bandwidth bottlenecks, and facilitates real-time alerting for enforcement dispatch.

Feasibility of Correlating Violations with Congestion Metrics

Quantifying the downstream congestion impact of an illegally parked vehicle requires sophisticated spatial-temporal data fusion. Research validates the feasibility of mathematically correlating stationary roadway occupation with traffic flow degradation. When a vehicle parks illegally on a carriageway, it initiates a physical conversion of movement space into static space. This phenomenon is empirically quantified using the Space Pressure Index (SPI), a composite metric derived from two critical dimensions: Longitudinal Occupation (OL) and Effective Width

Reduction (EWR)¹⁷. EWR measures the physical narrowing of the functional lane, while OL measures the cumulative equivalent length of parked vehicles against the segment length. Traffic engineering studies demonstrate a highly non-linear relationship between width reduction and capacity loss; for instance, field observations note that a 36.02% reduction in effective width can precipitate a 35.09% collapse in road capacity, severely degrading the Level of Service (LOS)²⁰.

By fusing CCTV-derived bounding box dimensions with fundamental traffic flow models, the system can dynamically estimate EWR. Furthermore, by integrating background traffic speeds

and queue length algorithms—defined as $QL = N \times L_v$, where N represents the

number of stopped vehicles and L_v is the average vehicle length—the AI can establish a direct causal link between a specific parking violation and the subsequent increase in the travel time index²¹. Previous attempts to model this fusion have successfully utilised microsimulation environments like SUMO (Simulation of Urban MObility) and PTV VISSIM to map the exact delay propagation caused by double-parking and curbside friction²².

Known Failure Modes and Limitations

Empirical data from existing deployments highlights several persistent failure modes that threaten algorithmic reliability. Occlusion is the most universally cited limitation for street-level optical systems; large transit buses or freight vehicles routinely block the camera's line of sight, hiding illegally parked cars in their visual shadow²⁵. Environmental degradation—comprising heavy rain, snow, fog, and night-time glare—drastically diminishes the efficacy of standard RGB cameras, causing false negatives in vehicle detection and catastrophic failures in Automatic Number Plate Recognition (ANPR)⁴.

High vehicle density exacerbates algorithmic confusion, frequently causing identity switches in tracking-by-detection heuristics like DeepSORT, leading the system to erroneously restart the dwell-time counter for a violating vehicle²⁸. Furthermore, acute camera angles can distort the perspective of parking boundaries, complicating the spatial logic required to determine if a vehicle is encroaching on a restricted zone. IoT ground sensors are not immune to failure; they are highly susceptible to physical destruction from municipal snowploughs and sweeping equipment, and they frequently suffer from magnetic interference in dense urban topologies⁷.

Regulatory and Privacy Feasibility

The deployment of continuous optical surveillance for automated enforcement intersects with stringent global privacy frameworks, notably the General Data Protection Regulation (GDPR) in Europe and the Digital Personal Data Protection (DPDP) Act in India. These regulations heavily restrict the indiscriminate capture, storage, and processing of personally identifiable information (PII), which legally encompasses vehicle license plates and the biometric facial data of pedestrians and drivers³⁰.

Traditional cloud architectures that archive raw, unredacted video streams pose massive compliance risks and invite severe regulatory penalties³⁰. Conversely, edge computing

architectures inherently facilitate "Privacy by Design." Because the raw video frames are analysed in volatile memory at the edge and immediately discarded, the system transmits only anonymised metadata and the isolated crop of the violating license plate, ensuring strict compliance¹⁶. Furthermore, to withstand legal scrutiny during the appeals process, automated systems must generate tamper-proof, cryptographically signed evidence packages containing timestamps, GPS coordinates, and clear visual proof of the infraction².

Cost Feasibility and Infrastructure Economics

The capital expenditure (CAPEX) required for deployment varies dramatically across technological approaches. Traditional magnetic or ultrasonic IoT sensors operate on a 1-to-1 ratio, necessitating the purchase, calibration, and invasive installation (core drilling) of a distinct sensor for every single monitored parking bay. Consequently, large-scale sensor deployments incur immense hardware and civil works costs¹⁶.

In stark contrast, modern AI-powered optical systems operate on a 1-to-many paradigm. A single high-definition edge-AI camera mounted on existing street furniture can monitor up to 125 parking spaces simultaneously. This architectural efficiency renders optical AI deployments up to three times cheaper than magnetic sensor arrays for equivalent spatial coverage¹⁶.

In the context of the Indian market, professional installation of commercial-grade CCTV networks ranges from ₹2,00,000 to ₹4,50,000 for mid-sized commercial deployments (15 to 30 cameras), while large-scale urban intersections requiring enterprise-grade PTZ cameras, structural cabling, and advanced analytics can scale from ₹8,00,000 to over ₹25,00,000³². Despite these upfront costs, the return on investment (ROI) is highly favourable. AI systems drastically reduce the need for constant human patrol units—which carry perpetual salaries, vehicle maintenance, and pension liabilities—often yielding a payback period of mere months due to automated revenue generation and reduced operational expenditure³⁵.

Verdict: Hackathon Scope vs. City-Scale Deployment

For a constrained 48-to-72-hour hackathon environment, physical hardware integration, custom edge-device quantisation, and multi-sensor IoT fusion are entirely out of scope. The most feasible pathway to demonstrating a functional prototype relies on orchestrating pre-existing, open-source models (e.g., YOLOv8 and EasyOCR) applied to publicly available traffic datasets (e.g., UA-DETRAC or PKLot). The hackathon team should concentrate strictly on the software logic: building a pipeline that processes an MP4 video stream, tracks vehicles entering designated restricted polygons, measures their dwell time, simulates a congestion impact score using a rudimentary EWR calculation, and triggers a mock API alert.

Conversely, executing a full city-scale deployment necessitates the procurement of ruggedised edge-inference nodes, integration with redundant 5G telecommunications backhaul, exact photometric calibration of cameras across varying urban topologies, secure API gateways linking to regional transport authority databases for automated fine generation, and rigorous adversarial testing against extreme weather and lighting conditions.

2. Feature Analysis

Core Module Decomposition and Algorithms

An enterprise-grade parking intelligence solution requires the seamless orchestration of multiple discrete analytical modules.

- **Detection Module:** This layer is responsible for vehicle localisation, illegal-parking spatial classification, and duration tracking. Modern systems overwhelmingly utilise the YOLO (You Only Look Once) family of single-stage detectors (v8, v9, v11) or SSD (Single Shot MultiBox Detector) due to their optimal equilibrium between mean Average Precision (mAP) and real-time processing speeds¹. Following spatial localisation, the system employs DeepSORT (Simple Online and Realtime Tracking with a Deep Association Metric) to assign and maintain unique identities across subsequent video frames, enabling the precise calculation of vehicular dwell time².
- **Spatial Mapping Module:** Translating pixel coordinates into global geolocations allows for macro-level analysis. To identify chronic violation zones rather than isolated incidents, the system applies density-based clustering. The ST-DBSCAN (Spatio-Temporal Density-Based Spatial Clustering of Applications with Noise) algorithm is uniquely suited for this task. Unlike standard DBSCAN, ST-DBSCAN evaluates both a spatial distance threshold (Eps_1) and a temporal distance threshold (Eps_2). This dual-parameter evaluation ensures that clustered hotspots represent areas with persistent, ongoing violations across time, rather than a fleeting anomaly¹¹.
- **Congestion Impact Quantification:** This critical module links static parking events to fluid traffic degradation. The system integrates fundamental traffic stream models, applying the formula $q = ku$ (where q is flow, k is density, and u is speed) alongside the Space Pressure Index to quantify capacity reduction¹⁷. Queueing theory and shockwave analysis are frequently employed to model how the frictional bottleneck caused by an illegally parked vehicle propagates delays upstream through the carriageway²¹.
- **Temporal Pattern Analysis:** To identify cyclical variations in parking behaviour, systems utilise Long Short-Term Memory (LSTM) networks or Quasi-Poisson regression models. These techniques analyse time-series data to detect periodic spikes correlated with commercial delivery hours, weekend recreational surges, or large-scale municipal events³⁹.
- **Enforcement Prioritisation Engine:** This scoring module fuses outputs from the preceding layers. Rather than dispatching officers based on geographical proximity, the engine assigns a dynamic priority score to each violation. This score is calculated by cross-multiplying the severity of the measured traffic flow disruption (e.g., delay in seconds) with the duration of the illegal parking event, ensuring resources are directed to the most disruptive infractions.
- **Alerting and Dispatch Mechanism:** The backend communication layer aggregates the required legal evidence—including an unredacted crop of the license plate, a wide-angle contextual image, GPS coordinates, and the priority score—and formats it into a JSON

payload. This payload is transmitted via secure REST APIs to the municipal e-challan or dispatch system³¹.

- **Dashboard Layer:** The user interface provides traffic wardens and urban planners with a Web-GIS visualisation, displaying real-time ST-DBSCAN heatmaps, active violation queues, and predictive congestion alerts¹¹.

Comparative Analysis: Commercial vs. Academic Solutions

A distinct philosophical divergence exists between off-the-shelf commercial enforcement products and the frameworks proposed in academic literature.

Commercial vendors (such as Hikvision, SenSen, and viAct) engineer their solutions with an intense focus on the detection, ANPR, and evidence-packaging modules³¹. Their primary value proposition is revenue assurance and operational automation. Consequently, their feature sets excel in edge-hardware optimisation, multi-sensor handoffs, and generating cryptographically secure, unassailable ticketing evidence. However, they treat the parking violation as an isolated static event; commercial systems are almost universally devoid of advanced traffic flow integration or congestion impact quantification.

Conversely, published academic research approaches the problem holistically, heavily emphasising the fluid dynamics of the urban environment. Researchers focus deeply on quantifying Effective Width Reduction, Level of Service (LOS) degradation, and microsimulation modelling using VISSIM and SUMO to prove how curbside friction degrades intersection throughput²².

The critical gap missing across the entire board is a unified platform that successfully merges the robust, tamper-proof ANPR enforcement pipelines of commercial vendors with the dynamic, real-time congestion quantification algorithms of academic research. Cities currently lack a system that not only issues the ticket but empirically explains the ticket's justification based on real-time traffic degradation.

3. Ideas for Novel and Advanced Features

Beyond the foundational requirements of detection and ticketing, an intelligent parking ecosystem can integrate advanced predictive, behavioural, and ethical features to shift the paradigm from reactive punishment to proactive management.

Predictive Forecasting

By leveraging historical ST-DBSCAN hotspot data alongside exogenous variables such as meteorological conditions, local event schedules, and commercial delivery manifests, the system can utilise Graph Convolutional Networks (GCN) to forecast where illegal parking will materialise before it occurs⁴⁷. Predicting demand surges allows municipal authorities to deploy temporary physical deterrents or preemptively station traffic wardens in high-probability zones, actively preventing the formation of congestion-inducing bottlenecks.

Citizen-Facing Integrations and Crowdsourcing

An adjacent feature set involves incorporating crowdsourced intelligence to bridge gaps in the

fixed CCTV infrastructure. Inspired by platforms reporting micromobility parking infractions, municipalities can deploy a dedicated mobile application allowing citizens to upload geo-tagged images of parking violations¹⁰. To mitigate spam and malicious reporting, the AI backend autonomously evaluates the uploaded image, authenticating the violation through object detection prior to human review¹².

Furthermore, the system's predictive occupancy data can be integrated into consumer navigation applications via open APIs. By transmitting real-time parking saturation nudges directly to Waze or Google Maps, the system can intercept drivers en route, warning them of destination saturation and actively rerouting them to available off-street municipal parking structures. This fundamentally eliminates the slow-moving "cruising" traffic that frequently devolves into illegal spillover parking.

Gamification and Behavioural Nudges

To stimulate civic participation in crowdsourcing platforms, the system can introduce gamification mechanisms. Verified citizen reports can be rewarded with "Violation Penalty Points" (VPC) or civic tokens, which could be redeemed for public transit credits or municipal tax discounts¹². Behavioural nudging can also be applied to the drivers; dynamic LED Variable Message Signs (VMS) installed near chronic hotspots can broadcast real-time warnings (e.g., "Illegal parking detected ahead; automatic fines active"), capitalising on the psychological deterrent of guaranteed enforcement.

Equity, Fairness, and Explainability

A profound ethical vulnerability of automated surveillance is the risk of systemic bias. Neighbourhoods with dense, historic urban planning often lack private driveways, forcing lower-income residents to rely heavily on contested on-street parking. Blanket automated enforcement can disproportionately target these demographics compared to affluent areas. An equity-balancing algorithm must be integrated to dynamically adjust enforcement thresholds or fine severities based on the objective availability of legal parking spaces within a defined radial catchment area.

Additionally, explainability features (XAI) are crucial for municipal transparency. Instead of outputting a generic violation code, a Large Language Model (LLM) integrated into the backend can generate a human-readable justification for the ticket: *"This vehicle was flagged for an illegal 18-minute dwell in a restricted commercial zone, resulting in a 36% reduction in effective carriageway width and causing an acute 4-minute delay at the downstream intersection."* This provides unassailable, highly contextualised evidence for administrative tribunals.

Adaptations from Adjacent Domains

Innovations from adjacent computer vision domains offer substantial precedent for extending system capabilities. The detection of illegal waste dumping relies heavily on trajectory analysis and temporal sequence modelling to identify when a vehicle stops, deposits an anomalous object, and departs²⁶. Similarly, land encroachment detection utilises semantic segmentation architectures, such as U-Net3+ and Mask-RCNN, to autonomously identify unauthorised

physical expansion beyond cadastral boundaries⁵¹. By applying these methodologies to the parking domain, the fixed CCTV infrastructure can simultaneously act as an environmental monitoring and structural encroachment grid without requiring supplementary hardware.

4. External Datasets

Training an AI-driven parking intelligence system requires vast amounts of annotated visual and spatial data. The following table exhaustively catalogues public datasets that provide essential ground-truth data for detection, OCR, and spatial correlation.

Dataset Name	Source / Host	Size	Format	Geographic Origin	License	Usage Note for Development
PKLot	Voxel51 / UFPR	12,416 images (~695k bounding boxes)	JPEG, XML (Polygons)	Brazil	CC BY 4.0	Crucial for training binary occupancy classifiers. Contains extreme weather variations (sunny, cloudy, rainy) ¹³ .
CNRPark-EXT	CNR / AIMH	~150,000 labelled image patches	JPEG	Italy	Academic Use	Expands upon standard datasets by including severe partial occlusions (trees, lampposts) and

						distinct camera angles ¹⁵ .
CCPD (Chinese City Parking)	GitHub / ECCV	~300,000 annotated images	JPEG (Annotations embedded in filename)	China	MIT	The industry standard for training highly robust ANPR/OCR pipelines. Includes challenging subsets for extreme blur, rotation, and tilt ²⁷ .
UA-DETRAC	Albany University	100 videos (140,000 frames)	MP4, XML (BBboxes)	China / USA	Non-Commercial	Vital for training Multi-Object Tracking (MOT) algorithms like DeepSORT. Annotates specific vehicle categories (car, bus, truck) in active

						traffic ⁵⁶ .
HighD	RWTH Aachen	147 hours (110k vehicles)	CSV (Trajectories), Video	Germany	Non-Commercial	A drone-captured dataset providing flawless, unoccluded vehicle trajectories. Essential for modelling queue lengths and fluid traffic degradation ⁵ .
DroneVehicle	GitHub / Tianjin Univ.	56,878 images	RGB & Infrared pairs	China	CC BY-NC-SA 4.0	Highly valuable for teams exploring cross-modality detection to overcome night-time and low-light environmental failure modes ⁴ .
CARPK	Kaggle	89,777	JPEG,	Taiwan	Open	Drone-vi

		bounding boxes	TXT (YOLO format)		Access	ew parking lot dataset ideal for training macro-level spatial density and capacity estimators ⁵⁹ .
NYC 311 Illegal Parking	NYC Open Data	Thousands of rows (Live streaming API)	CSV / JSON	New York, USA	Public Domain	Tabular data vital for building the spatial-temporal correlation engine, testing ST-DBSCAN clustering, and driving the dispatch dashboard ⁴¹ .

Identification of Data Gaps

Despite the sheer volume of available visual datasets, a critical operational gap exists: there is an absence of unified, synchronous multimodal datasets that perfectly pair street-level CCTV footage of illegal parking events with real-time, highly granular downstream traffic flow telemetry (such as ground-truth intersection delays or API speed drops). Hackathon teams cannot simply download a single file to prove that a specific parked car caused a specific delay. They will need to simulate this correlation computationally—by training an object detector on UA-DETRAC or PKLot, tracking the dwell time, and mathematically inferring the congestion

impact using the Space Pressure Index (SPI) applied to the extracted pixel dimensions.

5. Existing Notebooks, Models, and Reference Implementations

To accelerate prototype development, particularly under strict hackathon time constraints, teams must leverage existing open-source frameworks, tracking scripts, and pre-trained neural network weights.

Project / Model Title	Source Platform	Description & Architecture	Reusability vs. Need for Adaptation
benarnav/parking-monitor	GitHub	The reference implementation for the "Eye in the Sky" NYU research paper. Orchestrates YOLO object detection, analyses NYC DOT public camera feeds, and automates municipal 311 service request submissions ⁴¹ .	Extremely High. The modular architecture cleanly separates detection from API communication. Teams can fork this pipeline and immediately adapt it to local municipal APIs or generic webcam streams.
st_dbscan	GitHub / Python Package	A Python implementation of the ST-DBSCAN algorithm optimized with sparse matrices. Designed to cluster spatial-temporal movement data ³⁶ .	High. Easily installable via pip install st-dbscan. Can be dropped directly into the spatial mapping module to ingest bounding-box coordinates and output chronic hotspot clusters.
advanced-anpr-fr	GitHub	A comprehensive intelligent workflow utilising YOLOv8 for	Moderate to High. Provides an excellent reference

		vehicle localisation and character segmentation, explicitly designed to identify complex traffic violations and generate automated e-challans ² .	architecture for the backend database logging and OCR cropping, though its heavy reliance on facial recognition may require stripping out for strict parking use-cases.
Perception365/VehicleNet-Y26s	HuggingFace Hub	A multi-class vehicle detection model trained specifically on highly dense, heterogeneous Indian traffic (UVH-26-MV dataset) using the Ultralytics YOLO architecture. Capable of classifying 14 distinct vehicle types ⁶² .	High. Eliminates the need to train a vehicle detector from scratch. Its specialisation in heterogeneous, high-occlusion traffic environments makes it uniquely suited for urban developing-world applications.
nickmuchi/yolos-small-finetuned-license-plate-detection	HuggingFace Hub	A Vision Transformer (YOLOS) model fine-tuned specifically to detect and extract license plates, achieving a 49.0 AP on validation data ⁶⁴ .	High. Can be invoked instantly via the HuggingFace pipeline API. Ideal for bypassing the complex process of training a dedicated ANPR localiser.
prakharninja0927/illegal-parking-detection	GitHub	A legacy computer vision pipeline utilising YOLOv4 to highlight vehicles that violate	Moderate. The Region of Interest (ROI) mapping logic is highly reusable, but the core object

		predefined static parking coordinates within an MP4 video stream ⁶⁶ .	detection weights (YOLOv4) are outdated and must be replaced with modern v8 or v11 architectures to meet current inference speed standards.
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The Single Most Promising Starting Point

The undisputed optimum starting point for a hackathon team is the **benarnav/parking-monitor** repository⁴¹. While numerous GitHub repositories offer basic scripts that merely draw bounding boxes around cars, this repository embodies a complete, systems-level enforcement paradigm. It contains mature Python logic for establishing spatial Regions of Interest (ROIs), tracks continuous vehicle dwell time within those boundaries to definitively confirm an infraction, and includes the vital external communication logic required to interface with external dispatch APIs.

A competitive hackathon team can dramatically accelerate their development by forking this exact repository. By replacing the generic object detection model with the highly specialised VehicleNet-Y26s from HuggingFace, updating the OCR engine to handle local license plate fonts, and piping the output coordinates into the st_dbscan clustering library, a team can rapidly yield an enterprise-grade, congestion-aware prototype capable of dominating the competition within a 48-hour sprint.

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